

José María Barbeito

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PROFESSIONAL SUMMARY

Engineer with experience spanning semiconductor manufacturing, advanced materials, ballistic protection systems, software development, technical data systems, and aerospace research. Experienced in process development, statistical analysis, manufacturing optimization, technical prototyping, and building digital tools that support real-world engineering operations. Comfortable working across physical engineering, software, manufacturing, and product-development environments.

EDUCATION

North Carolina State University, *Raleigh, NC*

- **B.S.E:** Chemical Engineering; **Minor:** Material Science Engineering
- **Current Cumulative GPA:** 3.7/4.0

PROFESSIONAL EXPERIENCE

Regulatory and Standards Manager Delta V — Buenos Aires, Argentina

January 2026 – Current

- Supported engineering, product-development, and international expansion projects across ballistic protection, advanced materials, manufacturing systems, testing infrastructure, software, and technical business development.
- Developed and coordinated the end-to-end pathway for NIJ ballistic compliance/certification, including identification of applicable standards and threat levels, test planning, sample and product documentation, laboratory coordination, traceability requirements, test-data review, and preparation of the technical and quality systems required to support formal qualification.
- Supported the design and implementation planning for an independent ballistic testing facility, covering instrumentation, armor and projectile configurations, backface-deformation measurement, laboratory workflow, safety requirements, equipment sourcing, documentation, quality systems, and accreditation strategy.
- Contributed to planning for UHMWPE ballistic-material manufacturing, including process requirements, production equipment, technology transfer, supplier responsibilities, documentation, intellectual-property considerations, and manufacturing implementation.
- Participated in development of a proposed manufacturing operation in Angola, including on-site travel and facility/project assessment; evaluated production requirements, equipment, infrastructure, local operating constraints, logistics, implementation sequencing, and coordination with partners and stakeholders.
- Supported international project development and technical outreach across multiple markets, including industry engagement surrounding Eurosatory in Paris, identification and evaluation of suppliers and partners, technical discussions, commercial opportunities, and planning for future manufacturing and defense-sector projects.
- Designed and developed DeltaDash, a full-stack platform for organizing and analyzing ballistic test data, improving traceability across test sessions, individual shots, protocols, materials, locations, and supporting technical records.
- Evaluated photogrammetry and Gaussian Splatting workflows for detailed 3D documentation and visualization of facilities, products, and equipment, with applications in remote review, project planning, and technical presentation.
- Assisted with structuring manufacturing operations in Odoo, including product families, variants, Bills of Materials, and configuration logic, while working across engineering, management, laboratories, suppliers, logistics providers, and international partners to translate project requirements into executable technical plans.

Process Engineering Co-Op, Wolfspeed, Durham, NC

Summer 2023, 2024, Academic Year 2023-2024, 2024-2025

- Supported process engineering in high-volume SiC wafering and epitaxial manufacturing environments.
- Applied Statistical Process Control, Design of Experiments, Measurement Systems Analysis, and GR&R to identify process variation, optimize equipment, and improve manufacturing performance.
- Designed and tested proprietary process and tooling modifications intended to increase yield, reduce operating cost, and improve efficiency.
- Trained in operation and maintenance of Chemical Vapor Deposition equipment used for doped SiC epitaxial growth.

- Facilitated installation and qualification of FTIR metrology systems within Class 100 cleanrooms.
- Supported factory-layout planning, Process Change Review Boards, process qualification, and manufacturing-data analysis.

PROJECT EXPERIENCE

DeltaDash — Engineering Data Platform

- Designed and built a full-stack application consolidating fragmented ballistic-testing information into a structured engineering database and dashboard.
- Created workflows connecting test-level metadata with individual shot measurements, protocols, locations, materials, and technical records.
- Developed the application around actual engineering use cases including traceability, historical retrieval, comparison, and future analytics.
- Deployed frontend and backend services in production using Railway and worked through API, environment, configuration, and deployment issues.

Evolution Simulator — C++ / WebAssembly

- Developed a browser-based artificial-life and evolutionary simulation platform with a C++17 simulation core compiled to WebAssembly.
- Implemented an Entity-Component-System architecture supporting large numbers of independently simulated entities and extensible behavior systems.
- Integrated WebGPU rendering with WebGL2 fallback for hardware-accelerated browser visualization.
- Used the project to explore simulation architecture, procedural systems, emergent behavior, performance optimization, and complex-system design.

josebarbeito.com — Personal Technical Portfolio

- Designed and developed a personal portfolio showcasing engineering, software, photography, and independent R&D projects.
- Used the site as a production environment for frontend development, responsive design, media presentation, interactive content, and deployment.
- Structured technical work for communication to both engineering and nontechnical audiences.

Design and Manufacturing of Rotating Detonation Engine for Novel Power Generation *Academic Year 2024-2025*

- A member in a research team with NC State Liquid Rocketry Lab, focused on developing a turbine aimed at downstream power generation utilizing supersonic exhaust from a Rotating Detonation Engine (RDE).
- Developed control system that operated the turbine and handled data acquisition utilizing an in-house LabView program and a laboratory-grade suite of sensors, devices, and control equipment
- Achieved rudimentary fluid dynamic modeling of supersonic flow through turbine after detonation chamber to estimate thermodynamic models
- Received 3rd place in AIAA 2025 Region II Student Conference and named Best Chapter Award

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, TypeScript, SQL, HTML/CSS, JSL

Frameworks & Tools: Streamlit, Pandas, Plotly, OpenPyXL, Git/GitHub, WebAssembly, CMake, LabVIEW, Railway

AI & Development: AI-assisted software development, LLM workflows, rapid prototyping, debugging, architecture exploration, prompt engineering, agentic coding workflows

Product & Data: Requirements discovery, technical product design, data architecture, dashboard design, API integration, data pipelines, technical documentation, stakeholder communication

Engineering: SPC, DOE, MSA, GR&R, semiconductor manufacturing, metrology, process optimization

3D & Visualization: Gaussian Splatting, photogrammetry, 3D reconstruction, technical photography, drone imaging